Olivia Chiarini, she/they

linkedin.com/in/olivia-chiarini

Education

Drexel University Jan. 2025 – Present
B.S. in Physics GPA: 3.67
Transferred from **University of Maryland, Baltimore County**, Aug. 2022 – December 2024

Research Experience

Data Analysis and Run Control for the Scintillating Bubble Chamber Mar. 2026 – Present
Drexel University and Fermilab; PI: Russell Neilson, Ph.D.

- Worked in person at Fermilab on the SBC, including maintenance and data collection runs
- Updated documentation for access and analysis of SBC events
- Created reconstructions of events with bubble positions and size using multiple detector types including silicon photomultipliers and piezoelectrics
- Developed algorithm for bubble multiplicity calculation and visualization
- Led handscanning campaign for tens of thousands of nucleation events

Astronomical Data Reduction Pipeline Aug. 2023 – Jan. 2025
University of Maryland, Baltimore County (UMBC); PI: Eileen Meyer, Ph.D.

- Developed an astronomical data reduction pipeline
- Benchmarked said pipeline against the Landolt Standard Stars as a part of a UMBC Undergraduate Research Award
- Developed data analysis pipeline to determine the FWHM of astronomical sources and aid in the characterization of the atmosphere around UMBC

Execution of Density Functional Theory Workflows on HPC Cluster Jun. 2024 – Aug. 2024
SCIART Summer Program, UMBC; PI: Joseph Bennet, Ph.D.

- Used UMBC's High Performance Computing Cluster for Density Functional Theory research as part of Andrew W. Mellon Foundation's SCIART Internship
- Analyzed the usage of Titanium Carbide MXenes as films to protect against harmful adsorptions
- Presented poster and abstract at UMBC's Summer of Undergraduate Research Festival

Simulating the AXIS Space Telescope Observation Scheduler Mar. 2023 – Sep. 2023
UMBC; PI: Eileen Meyer, Ph.D.

- Optimized computer simulations for satellite scheduling
- Developed procedures for scheduled observations and recovery from time-sensitive interruptions

Professional Experience

Teaching Assistant May 2025 – Present
Drexel University

- Graded homework, labs, exams, and quizzes for introductory physics classes
- Supported and ran lab classes for introductory physics classes in mechanics, electricity & magnetism, thermodynamics, and optics
- Supported and ran discussion sections for small group problem solving
- Wrote, proctored, and distributed exams for large groups of physics students

Senior Telescope Operator and Research Analyst Nov. 2023 – Dec. 2024
University of Maryland, Baltimore County

- Created the current data reduction pipeline
- Delivered and discussed physics demonstrations to large audiences
- Wrote and delivered short talks on topics in astronomy to large audiences
- Developed astronomical imaging pipelines for processing astronomical imaging using Python, acted as software maintainer in GitHub by fixing bugs and adding requested features

Olivia Chiarini, she/they

linkedin.com/in/olivia-chiarini

i3b's The Pride Center Student Supervisor Aug. 2022 – May. 2024
University of Maryland, Baltimore County

- Planned and hosted social and educational events/trainings for over 500 unique students and staff.
- Helped organize and process data for student engagement in MS Excel.
- Supervised, managed, and trained student staff interns throughout i3b on event planning, conflict resolution, and communication.
- Organized regular meetings with professional and student staff, planning agendas and running said meetings as necessary.
- Worked to engage directly with undergraduates, graduate students, faculty, and staff.

Technical and Academic Skills

- **Physics:** Observational Astronomy, Galactic Astrophysics, Cosmology, General Relativity, Neutrino Physics, Particle Physics, Statistical Mechanics, Electricity & Magnetism, Quantum Mechanics, Classical Mechanics, Optics w/Lab, Electronics w/Lab
- **Math:** Linear Algebra, Differential Equations, Multivariable Calculus, Differential Geometry, Tensor Calculus
- **Computing:** Comfortable with a variety of operating systems: Windows, macOS, Ubuntu, Pop_OS, Arch. Experience programming in Python, C, C++, C#, Java, and Bash. Algorithm optimization with IBM CPLEX Engine. Proficient in MS Office Suite, Google ecosystem, and LaTeX.
- **Logistics & Planning**: Comfortable working with a variety of institutional partners to plan and execute large scale events (300–350 people).

Presentations and Publications

- Chiarini, O. (2026, November 13–15). Scintillating Bubble Chamber Liquid Argon 10 kg (SBC-LAr10) Detector Analysis [Conference presentation]. APS Mid-Atlantic Section Annual Meeting 2026, Baltimore, MD, United States
- Chiarini, O., Doherty, C., Bennett, J., Rosenzweig, Z. 2-DM (2024, August 7th) MXenes as Chemical Sensors in the art museum environment [Conference presentation]. UMBC's Summer Undergraduate Research Fest, Baltimore MD, United States

Leadership, Accolades, & Volunteer Experience

- The Susan and Donald Larson Endowed Scholarship, 2026
- Barry Goldwater Scholarship and Excellence in Education Foundation Finalist, 2026
- Drexel Kaczmarczik Day Volunteer 2025, 2026
- Drexel Dean's List Winter 2025, Spring 2025, Summer 2025, Fall 2025
- A. J. Drexel Scholarship 2025, 2026
- UMBC Undergraduate Research Award, 2024
- Vice President, UMBC Astronomy Club 2024-2025
- Delivered outreach at Astronomy Festival on the National Mall June 22, 2024
- Andrew W. Mellon Foundation SCIART Internship, Summer 2024
- Proctored 2024 Science Olympiad Division C Invitational Astronomy Exam Spring 2024
- Recipient of State Delegate Kathy Forbes Scholarship 2022, 2023, 2024
- Recipient of State Senator Chris West Scholarship 2022, 2023, 2024
- Recipient of George Washington Carver Center IT/IMP Scholarship 2022